

OR Multiple-Choice Bank + Answers

Multiple-choice bank

Exam format: "Mark all correct answers (there can be more than one)." All correct = **2 pts** · one deviation = **1 pt** · two or more, or blank = **0**.

Answers are in [mc-answers.md](#) — separate file, so this one is a real test. Items marked **[SS24]** / **[SS25]** are from real papers.

§1 · LP basics and simplex

Q1. Which are correct about converting to standard form?

- ☐ a $\leq$ constraint gets a slack variable, and that slack column can serve as a basis column
- ☐ a $\geq$ constraint gets a surplus variable whose column can serve as a basis column
- ☐ a free variable is replaced by $x = x^+ - x^-$ with $x^+, x^- \geq 0$
- ☐ $\min c^T x$ and $\max c^T x$ have the same optimal solution set

Q2. A point v of $X = \{x : Ax \leq b, x \geq 0\}$ is a vertex if and only if...

- ☐ it has n linearly independent active constraints
- ☐ it corresponds to a basic feasible solution
- ☐ it is not a convex combination of two other points of X
- ☐ it maximises some linear objective over X

Q3. Which can happen for a linear program?

- ☐ the feasible region is empty
- ☐ the feasible region is non-empty and the objective is unbounded
- ☐ there are exactly two optimal solutions
- ☐ there are infinitely many optimal solutions

Q4. [SS25] Which scenarios can cause the simplex algorithm to cycle?

- ☐ the simplex algorithm never cycles
- ☐ the feasible region is unbounded
- ☐ there are multiple distinct optimal solutions
- ☐ the problem is degenerate

Q5. In a max simplex tableau (z-row holding $-c$), which readings are correct?

- ☐ no negative z-row entry $\Rightarrow$ the current basis is optimal
 - ☐ the entering column has no strictly positive entry $\Rightarrow$ the LP is unbounded
 - ☐ a tie in the ratio test $\Rightarrow$ the next basis is degenerate
 - ☐ a basic variable equal to zero $\Rightarrow$ the LP is infeasible
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§2–3 • Tableau reading and sensitivity

Q6. From an optimal tableau you can read off directly...

- ☐ the values of the basic variables
- ☐ the shadow prices
- ☐ the optimal objective value
- ☐ the number of simplex iterations that were performed

Q7. About shadow prices y in a max problem:

- ☐ $y^T = c_B^T B^{-1}$
- ☐ they appear in Row 0 under the slack variables
- ☐ a constraint with positive slack has shadow price zero
- ☐ y_i remains valid for arbitrarily large changes in b_i

Q8. [SS25] You perturb the RHS of a constraint by Δ and the basis stays optimal. Which are always true?

- ☐ the shadow price vector π should be updated
- ☐ π can be used to find the change in the values of the basic variables
- ☐ π can be used to find the change in the objective value
- ☐ the solution can become degenerate for some Δ

Q9. Cost ranging. Which are correct?

- ☐ changing a **non-basic** variable's cost can change the optimal solution but not the objective value, while it stays in range
- ☐ changing a **basic** variable's cost changes the objective value but not the plan, while it stays in range
- ☐ a basic variable's cost range is computed from that variable's **row** of the tableau
- ☐ a non-basic variable's cost range is two-sided

Q10. In an optimal tableau:

- ☐ a basic variable with RHS zero indicates degeneracy
- ☐ a non-basic variable with Row-0 entry zero indicates multiple optimal solutions
- ☐ degeneracy and multiple optima are the same phenomenon
- ☐ the endpoints of a RHS validity range are exactly where the basis becomes degenerate

Q11. A new product needs resources a_{new} and sells at c_{new} . It is worth introducing iff...

- ☐ $c_{\text{new}} > y^T a_{\text{new}}$
- ☐ its Row-0 entry $y^T a_{\text{new}} - c_{\text{new}}$ is negative
- ☐ a_{new} uses less of every resource than some existing product
- ☐ the LP must be re-solved from scratch to decide

§4 • Duality

Q12. [SS25] Which are correct?

- ☐ every linear program has a dual program
- ☐ if a primal constraint is not tight, its corresponding dual variable is zero
- ☐ if the primal is infeasible, the dual is unbounded
- ☐ if the primal is unbounded, the dual is infeasible

Q13. About the dual of $\max c^T x$ s.t. $Ax \leq b, x \geq 0$:

- ☐ the dual has one variable per primal constraint
- ☐ the dual has one constraint per primal variable
- ☐ the sign restrictions $x \geq 0$ each contribute a dual variable
- ☐ the dual of the dual is the primal

Q14. In a **max** primal, which SOB classifications are right?

- ☐ a $\geq$ constraint is bizarre, so its dual variable satisfies $y \leq 0$
- ☐ an $=$ constraint is odd, so its dual variable is free
- ☐ a free primal variable gives a dual **equality** constraint
- ☐ a primal variable $x \leq 0$ gives a dual constraint in the sensible direction

Q15. Weak and strong duality:

- ☐ $c^T x \leq b^T y$ holds for every feasible x and every feasible y
- ☐ at optimality $c^T x^* = b^T y^*$
- ☐ if both problems are feasible, both have finite optima
- ☐ a duality gap can exist for a feasible bounded LP

Q16. To show a given feasible x^* is **not** optimal using complementary slackness, you...

- ☐ solve the primal LP and compare objective values
- ☐ set $y_i = 0$ for every slack primal constraint
- ☐ force dual constraint j to equality for every $x_j \neq 0$
- ☐ check the recovered y against the dual's remaining constraints and sign restrictions

§5 · IP modelling and complexity

Q17. About the LP relaxation of a maximisation IP:

- ☐ its optimal value is an upper bound on the IP optimum
- ☐ if its optimum is integral, it solves the IP
- ☐ rounding its optimum always gives a feasible integer solution
- ☐ with integer objective coefficients you may round the bound down to $\lfloor z_{LP} \rfloor$

Q18. Two correct formulations of the same IP. The **stronger** one is the one whose...

- ☐ LP relaxation has a smaller feasible region
- ☐ integer feasible set is smaller
- ☐ relaxation gives a tighter bound, so branch & bound prunes earlier

- ☐ constraint count is smaller

Q19. Big-M constraints:

- ☐ M must be a constant, not a decision variable
- ☐ choosing M too small can cut off feasible solutions
- ☐ choosing M very large weakens the LP relaxation
- ☐ $t = 1 \Leftrightarrow x > K$ can be modelled exactly for a continuous x

Q20. Complexity:

- ☐ deciding whether an integer program has a feasible solution is NP-hard
- ☐ linear programming is in P
- ☐ the simplex algorithm runs in polynomial time in the worst case
- ☐ every NP-hard problem is in NP

Q21. Which are on Karp's list of NP-complete problems as presented in the lecture?

- ☐ the travelling salesperson problem
 - ☐ knapsack
 - ☐ set cover
 - ☐ linear programming
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§6 · Branch and bound and cuts

Q22. In branch & bound on a **maximisation** problem, a node may be pruned when...

- ☐ its relaxation is infeasible
- ☐ its relaxation optimum is integral
- ☐ its relaxation value is $\leq$ the incumbent
- ☐ its relaxation value is $\geq$ the incumbent

Q23. Branching on a fractional $x_i = 2.25$ into $x_i \leq 2$ and $x_i \geq 3$:

- ☐ discards no integer feasible point
- ☐ removes the current fractional optimum from both children
- ☐ each child inherits all constraints added by its ancestors
- ☐ the two children have disjoint LP feasible regions

Q24. [SS25] About cutting planes:

- ☐ they remove integer solutions from the LP relaxation's feasible region
- ☐ an ideal outcome is the convex hull of all integer feasible solutions
- ☐ the method always terminates with exactly two Gomory cuts
- ☐ combining cuts with branch & bound can speed up solving integer programs

Q25. Node-selection strategies:

- ☐ FIFO is breadth-first and takes the oldest open node

- ☐ LIFO is depth-first and takes the newest open node
 - ☐ LIFO tends to find a first incumbent sooner
 - ☐ FIFO always explores fewer nodes than LIFO
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§7 · Total unimodularity

Q26. A is totally unimodular. Which follow?

- ☐ every entry of A lies in $\{-1, 0, +1\}$
- ☐ A^T is totally unimodular
- ☐ $[A, I]$ is totally unimodular
- ☐ every square submatrix of A is invertible

Q27. [SS25] A is TU and $b \in \mathbb{Z}^m$. Consider $\max\{c^T x : Ax \leq b\}$. Which claims are correct?

- ☐ if c is integral, every optimal vertex solution x^* has integral value $c^T x^*$
- ☐ adding the constraint $x \geq 0$ can destroy total unimodularity of A
- ☐ there exists a bipartite graph G such that A is the incidence matrix of G
- ☐ the determinant of every submatrix of A is in $\{+1, -1, 0\}$

Q28. Which matrices are guaranteed TU?

- ☐ the incidence matrix of a bipartite graph
 - ☐ the incidence matrix of a directed graph
 - ☐ the incidence matrix of any undirected graph
 - ☐ a $\{0, 1\}$ matrix with the consecutive-ones property
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§8 · Matroids

Q29. [SS25] Let E be a finite set. For which collections $\mathcal{I}$ is $(E, \mathcal{I})$ a matroid?

- ☐ $\mathcal{I} = \{S \subseteq E : |S| \text{ is even}\}$
- ☐ $\mathcal{I} = \{S : S \subseteq E\}$
- ☐ $\mathcal{I} = \{S \subseteq E : S \text{ contains no cycle}\}$, E the edge set of a graph
- ☐ $\mathcal{I} = \{S \subseteq E : |S| \text{ is odd}\}$

Q30. The exchange axiom says: for $A, B \in \mathcal{I}$ with $|A| < |B|$...

- ☐ there is an $x \in B \setminus A$ with $A \cup \{x\} \in \mathcal{I}$
- ☐ there is an $x \in A \setminus B$ with $B \cup \{x\} \in \mathcal{I}$
- ☐ the smaller set is the one that grows
- ☐ the new element comes from the larger set

Q31. In a matroid:

- ☐ all bases have the same cardinality
- ☐ a basis is a maximal independent set

- ☐ the greedy algorithm in decreasing weight order returns a maximum-weight basis
 - ☐ the intersection of two matroids on the same ground set is a matroid
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§9 · Knapsack and approximation

Q32. The knapsack DP with table $B[i, w]$:

- ☐ runs in $O(n \cdot W)$ when indexed by weight
- ☐ runs in polynomial time in the input length
- ☐ can be indexed by value instead, in $O(n \cdot V)$
- ☐ recovers the chosen items by backtracking through the table

Q33. Approximation classes:

- ☐ $P \subseteq \text{FPTAS} \subseteq \text{PTAS} \subseteq \text{APX}$
 - ☐ knapsack admits an FPTAS
 - ☐ an FPTAS runs in time polynomial in n and in $1/\epsilon$
 - ☐ every NP-hard optimisation problem admits a constant-factor approximation
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§10 · Network flow

Q34. For an s - t cut $S = [X, V \setminus X]$:

- ☐ its capacity counts arcs from X to $V \setminus X$ only
- ☐ arcs from $V \setminus X$ back to X also contribute to the capacity
- ☐ every flow value is at most every cut capacity
- ☐ the minimum cut is unique

Q35. Ford–Fulkerson and the residual network:

- ☐ a forward residual arc has capacity $u(e) - f(e)$
- ☐ a backward residual arc has capacity $f(e)$
- ☐ backward arcs let the algorithm undo an earlier routing decision
- ☐ without backward arcs, greedy path-finding still finds the maximum flow

Q36. Which are true?

- ☐ max-flow equals min-cut
 - ☐ with integral capacities there is an integral maximum flow
 - ☐ with all capacities odd, some maximum flow has $f(e)$ odd on every arc
 - ☐ the min cut can be read off as the nodes reachable from s in the final residual network
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§11 · TSP

Q37. Definitions:

- ☐ an Eulerian cycle visits every edge exactly once
- ☐ a Hamiltonian cycle visits every node exactly once
- ☐ deciding whether a graph is Eulerian is NP-complete
- ☐ TSP asks for a minimum-weight Hamiltonian cycle

Q38. Subtour elimination:

- ☐ the degree constraints alone already forbid subtours
- ☐ SEC has exponentially many constraints
- ☐ MTZ has $O(n^2)$ constraints
- ☐ MTZ gives a weaker LP relaxation than SEC

Q39. Approximation algorithms for metric TSP:

- ☐ MST-doubling is a 2-approximation
- ☐ Christofides is a $3/2$ -approximation
- ☐ Christofides adds a minimum-weight perfect matching on the odd-degree vertices of the MST
- ☐ the nearest-neighbour heuristic has a constant-factor guarantee

§12–14 • Nonlinear

Q40. At a critical point with Hessian H (2×2 , $\det = ad - b^2$, $\text{tr} = a + d$):

- ☐ $\det > 0$ and $\text{tr} > 0 \Rightarrow$ local minimum
- ☐ $\det < 0 \Rightarrow$ saddle point
- ☐ $\det = 0 \Rightarrow$ the point is a saddle
- ☐ $\det > 0$ and $\text{tr} < 0 \Rightarrow$ local maximum

Q41. Leading principal minors of a Hessian:

- ☐ all $D_k > 0 \Rightarrow$ positive definite
- ☐ all $D_k < 0 \Rightarrow$ negative definite
- ☐ $D_1 < 0, D_2 > 0, D_3 < 0, \dots \Rightarrow$ negative definite
- ☐ some $D_k = 0 \Rightarrow$ the criterion does not apply and you should use eigenvalues

Q42. Convexity:

- ☐ $H_f \geq 0$ on a convex domain $\Leftrightarrow f$ is convex
- ☐ $H_f > 0$ everywhere $\Rightarrow f$ is strictly convex
- ☐ f strictly convex $\Rightarrow H_f > 0$ everywhere
- ☐ a convex function on a convex feasible set has every local minimum global

Q43. Convex sets:

- ☐ the circle $\{x : \|x\| = r\}$ is convex
- ☐ the closed disk $\{x : \|x\| \leq r\}$ is convex
- ☐ the intersection of two convex sets is convex
- ☐ the union of two convex sets is convex

Q44. Compactness:

- ☐ in $\mathbb{R}^n$, compact $\Leftrightarrow$ closed and bounded
- ☐ $[0, \infty)$ is closed but not bounded
- ☐ a compact non-empty set plus a continuous f guarantees a global min and max
- ☐ "not open" is the same as "closed"

Q45. [SS25] Which are true?

- ☐ gradient descent always converges to the global minimum for strictly convex functions, independent of step size
- ☐ gradient methods with momentum always need fewer iterations than without
- ☐ for a constrained problem, a KKT point is always a local optimum
- ☐ for a constrained **convex** problem a KKT point is always a global minimiser if Slater's condition holds

Q46. In the KKT system for $\min f$ s.t. $g_i \leq 0$, $h_j = 0$:

- ☐ the inequality multipliers satisfy $\lambda_i \geq 0$
- ☐ the equality multipliers μ_j are free in sign
- ☐ complementary slackness says $\lambda_i \cdot g_i(x^*) = 0$
- ☐ an inactive constraint may have a strictly positive multiplier

Q47. Constraint qualifications:

- ☐ Slater requires a point that satisfies every inequality **strictly**
- ☐ Slater requires convexity of f and the g_i
- ☐ under LICQ, a KKT point is guaranteed to be a global optimum
- ☐ under LICQ you must compare objective values across the candidates

Self-check

Twenty minutes, no notes, before the exam. Score yourself with the real rule: **all correct 2 · one deviation 1 · otherwise 0.**

If you drop points on	Reread
Q1–Q5	card §1
Q6–Q11	card §2, §3
Q12–Q16	card §4
Q17–Q21	card §5
Q22–Q25	card §6
Q26–Q28	card §7
Q29–Q31	card §8
Q32–Q33	card §9
Q34–Q36	card §10
Q37–Q39	card §11
Q40–Q47	card §12, §13, §14

Multiple-choice bank — answers

Questions are in [mc-question-bank.md](#). Score yourself with the real rule: **all correct 2 · one deviation 1 · otherwise 0**.

§1 · LP basics and simplex

Q1 — 1, 3. 2 ✗ — the surplus column is -1 , not $+1$, so it is *not* a basis column; that's why artificial variables exist. 4 ✗ — $\min c^T x = -\max(-c^T x)$; you negate the objective, not reuse it.

Q2 — 1, 2, 3. 4 ✗ — true for *some* objective only if the LP is bounded in that direction, and non-vertex points can also be maximisers when there are multiple optima.

Q3 — 1, 2, 4. 3 ✗ — if two distinct points are optimal, every convex combination of them is too, so you get infinitely many. Never exactly two.

Q4 — 4 only. Cycling requires a degenerate pivot — a basis change with no objective improvement.

Q5 — 1, 2, 3. 4 ✗ — that's degeneracy, and a degenerate BFS is perfectly feasible.

§2–3 · Tableau reading and sensitivity

Q6 — 1, 2, 3. The tableau records the final state only, not the path taken.

Q7 — 1, 2, 3. 4 ✗ — valid only while the basis stays optimal, i.e. inside the RHS range.

Q8 — 3, 4. 1 ✗ — if the basis is unchanged, so is π . 2 ✗ — the basic variables move by $B^{-1}\Delta b$, not by π . π gives only the *objective* change.

Q9 — 2, 3. 1 ✗ — inside the range *neither* the plan nor z changes; you're making none of it. 4 ✗ — one-sided: $c_j \leq y^T a_j$.

Q10 — 1, 2, 4. 3 ✗ — degeneracy is one point described by several bases; multiple optima is several points with the same z .

Q11 — 1, 2 — the same statement twice. 3 ✗ — irrelevant. 4 ✗ — that's the whole point of pricing a column.

§4 · Duality

Q12 — 1, 2, 4. 3 ✗ — infeasible primal $\Rightarrow$ dual is unbounded **or** infeasible. Both-infeasible is a real case.

Q13 — 1, 2, 4. 3 ✗ — sign restrictions are not counted as constraints.

Q14 — 1, 2, 3. 4 ✗ — $x \leq 0$ is bizarre, so its dual constraint takes the *bizarre* direction.

Q15 — 1, 2, 3. 4 ✗ — LPs have no duality gap; that's strong duality. Gaps are an *integer* programming phenomenon.

Q16 — 2, 3, 4. 1 ✗ — the entire point is that you never solve the LP.

§5 · IP modelling and complexity

Q17 — 1, 2, 4. 3 ✗ — rounding frequently lands outside the feasible region.

Q18 — 1, 3. 2 ✗ — the integer sets are identical; that's what "same IP" means. 4 ✗ — the disaggregated linking form has *more* constraints and is stronger.

Q19 — 1, 2, 3. 4 ✗ — $\{x > K\}$ is not closed, so it isn't a polytope. Only for integer x .

Q20 — 1, 2. 3 ✗ — Klee–Minty cubes force 2^n vertices. LP is in P via ellipsoid/interior-point, not simplex. 4 ✗ — NP-hard does not require membership in NP; NP-**complete** does.

Q21 — 1, 2, 3. 4 ✗ — LP is in P.

§6 · Branch and bound and cuts

Q22 — 1, 2, 3. 4 ✗ — that's the *minimisation* rule. A node above the incumbent is exactly the one worth exploring.

Q23 — 1, 2, 3, 4. All four hold — no integer lies strictly between 2 and 3, and the two half-spaces don't overlap.

Q24 — 2, 4. 1 ✗ — by definition a valid inequality never removes an integer point. 3 ✗ — no such guarantee.

Q25 — 1, 2, 3. 4 ✗ — depends entirely on the instance. On SS25's tree FIFO happened to be shorter; that's not a rule.

§7 · Total unimodularity

Q26 — 1, 2, 3. 4 ✗ — determinant 0 is allowed, and that means singular.

Q27 — 1, 4. 2 ✗ — appending an identity block preserves TU. 3 ✗ — bipartite incidence matrices are TU, but not every TU matrix is one.

Q28 — 1, 2, 4. 3 ✗ — an odd cycle breaks it; the graph must be bipartite.

§8 · Matroids

Q29 — 2, 3. 1 ✗ — not hereditary: $\{a, b\} \in I$ but $\{a\} \notin I$. 4 ✗ — $\emptyset \notin I$, since $|\emptyset| = 0$ is even. Fails axiom 1.

Q30 — 1, 3, 4. 2 ✗ — that's the direction reversed, the standard error.

Q31 — 1, 2, 3. 4 X — it's an independence system, but generally not a matroid. That's exercise T8.1.

§9 · Knapsack and approximation

Q32 — 1, 3, 4. 2 X — pseudopolynomial. W is a number; writing it takes $\log W$ bits.

Q33 — 1, 2, 3. 4 X — max clique, set cover and max independent set are not in APX at all.

§10 · Network flow

Q34 — 1, 3. 2 X — backward arcs contribute nothing; they cannot carry anything toward t . 4 X — the *value* is unique, the cut achieving it need not be.

Q35 — 1, 2, 3. 4 X — greedy alone can get stuck strictly below the maximum.

Q36 — 1, 2, 4. 3 X — this is SS25 E6a's counterexample: two odd flows merging give an even one.

§11 · TSP

Q37 — 1, 2, 4. 3 X — Eulerian is easy (check degrees and connectivity). *Hamiltonian* is the hard one.

Q38 — 2, 3, 4. 1 X — degree constraints alone are the assignment problem, which permits disjoint subtours.

Q39 — 1, 2, 3. 4 X — nearest neighbour has no guarantee at all.

§12–14 · Nonlinear

Q40 — 1, 2, 4. 3 X — $\det = 0$ means the test is **inconclusive**, not that it's a saddle. Go to eigenvalues.

Q41 — 1, 3, 4. 2 X — the pattern **alternates**, starting negative. Check against $-I$.

Q42 — 1, 2, 4. 3 X — $f(x) = x^4$ is strictly convex but $f'(0) = 0$.

Q43 — 2, 3. 1 X — the chord between two points on a circle passes through the interior, which isn't on the curve. 4 X — two disjoint disks give an obvious counterexample.

Q44 — 1, 2, 3. 4 X — $[0, 1)$ is neither; $\mathbb{R}^n$ is both.

Q45 — 4 **only**. 1 X — too large a step diverges. 2 X — "always" is false. 3 X — without a CQ, KKT points need not be optima.

Q46 — 1, 2, 3. 4 X — inactive means $g_i < 0$, so complementary slackness forces $\lambda_i = 0$.

Q47 — 1, 2, 4. 3 X — that's Slater's guarantee. LICQ yields candidates only.